

## SECTION 2.5: INFORMATION MAPPING

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## Valuing Information as a Strategic Asset

Information is now recognized as having real value. As an enterprise asset, information must be managed and cultivated as part of an overall enterprise strategy. In a *knowledge economy*, information is one of the most valued enterprise assets. In a *transformation economy*, proper governance of information becomes a *critical success factor*. Data and information will outlast applications, requiring the management of data and knowledge assets through time as applications, and even business processes change. (NASCIO: *Data Governance – Managing Information As An Enterprise Asset Part I – An Introduction*)

Accurate, timely, relevant information is crucial to good decision-making, including strategic decisions. Information and knowledge are key assets in the current knowledge worker driven economy. It has been consistently shown that information is essential for innovation, together with a culture that encourages and rewards intelligent risk taking. Information facilitates the assessment of both upside and downside risk associated with a course of action.

In the knowledge economy, successful organizations recognize that learning takes place at all levels. Learning organizations are embracing big data to extend their collective knowledge using ‘pervasive business intelligence’, which describes the use of information resources to support decision-making throughout the organization at all levels). (*Valuing Information as an Asset*).

Information is owned by business stakeholders; internal (functional areas, LOB, supply chain) and external (suppliers, partners, and customers). Business stakeholders provide operational guidance for information they own directly and information that is shared across one or more sets of stakeholders.

There is a set of business rules that are reliant on the data objects and data quality dimensions (see list below) to ensure the business process deliver the expected outputs. These outputs can impact other processes and ultimately capabilities used to support value streams. The value of a given information asset depends on the number of business rule sets, processes, and capabilities that are impacted. The value of an information asset can quickly expand if it is combined with other information assets and reused to support other business rule sets.

These relationships are explored using information mapping. Before embarking on information mapping, it’s important to review the following principles used to guide the process to ensure an optimum result.

## **Principles of Information Mapping**

### **Information must be bounded by the organization's vocabulary**

Common, shared business vocabulary streamlines collaboration, communication and automation. Stakeholder and internal communication and collaboration are often compromised when business vocabulary is not consistently defined. Simple terms such as customer, account, and product often have many meanings across business units and stakeholder environments. This may be fine in general conversation, but when it results in misstating financials, miscommunicating requirements, or deploying solutions that miss the mark, it can be costly.

### **Business concepts are bounded by the BA metamodel**

Business concepts are higher-level abstractions of business objects and data objects. These are sometimes referred to as “subject areas” and are commonly derived and/or extracted from the business architecture metamodel (see BIZBOK section 5.1 for getting started). The BA MM is a good starting point, but it may not be complete since there may be additional business concepts that are identified when defining the Capability Map (see BIZBOK section ??) or through business process improvement initiatives. The BA MM informs and can be informed by the Capability Map and similar recursive relationships exist when the organization introduces and/or updates their models for processes, information architecture, application architecture, and/or adopts existing metamodels from OMG.

### **Business Concepts are used to form a Conceptual View of the business**

Business concepts are organized into a logical representation depicting their relationships to each other to help stakeholders (including architects) navigate the organization's business model. These relationships can be derived from the BA MM (depending on the level of maturity and how far it has been extended) and from the capability maps and high-level business process models (if either exist). This “map” serves as the business reference model for creating the Logical Models (used to develop and/or improve the information architecture).

### **Information Objects are derived from and inform the enterprise metadata**

Information objects are logical representations of real-world objects (order, inventory, customer, vendor, ...). While business concepts are derived using a top-down approach, information objects are typically extracted / identified using a bottom-up approach. Information objects are used to support the information architecture, application architecture, and business processes. These objects can also be found in published metamodels (see OMG publications relevant for functional / vertical needs). There are also numerous ISO standards,

which can be leveraged (these contain taxonomies, terms, and information objects which support interoperability).

### **Information Objects are categorized and characterized using Logical Models**

They are often represented (organized) in taxonomies (which provide the baseline for categorizing information objects) and axioms (which constrain the number of possible interpretations / instances of objects within a taxonomy). These taxonomies are assembled into ontologies, which provide the building blocks for designing the logical and physical structures of data, information, and knowledge across the information architecture. No data objects exist outside of ontologies and all data objects need to align with business concepts. Data objects inherit data attributes from their parent objects, all of which are contained in the enterprise metadata model.

### **Business rules are intrinsically associated with business information**

These rules are important for the consistency of the information and must accompany the information whenever and wherever it is used. For example, an owner of a bank account must provide an address for notices from the bank. In legal terms, the bank has fulfilled its obligation to notify the customer by mailing a notice to the listed address. Without an address, the bank has no way to satisfy this obligation – hence the business rule requirement that an address must exist for each account holder, and, in many cases, that the account holder is solicited to update this address whenever contacting the bank.

The business rules are also used to identify the explicit and implicit nature of relationships between business and information objects and amongst information objects and their attributes. Discovery of these rules is an organic process as scenarios are identified in the course of ongoing project work.

### **How to Do Information Mapping**

This section discusses the procedures for defining the information maps, which provide the baseline for designing and improving the information architecture for an organization. The procedures are presented in a progression, but the order is not prescriptive as organizations may revisit most (if not all) of these procedures during end-2-end transformation efforts or selectively as part of an initiative. This mapping only provides the baseline for guiding the creation and/or extension of an organization's metadata and its information architecture. It is not intended to address all of the complexities associated with information architecture.

## **Capture the Business Vocabulary (Information Terms and Definitions)**

The business vocabulary represents the terms, meanings, and semantics used throughout the organization. Depending on the level of maturity and availability of business and information architecture artifacts, there is any number of starting points from which to create or extend the business vocabulary. Business terms and semantics can be extracted from business architecture metamodel (BA MM), enterprise metadata and glossary, capability maps, business process maps, information assets (systems, documentation), operational documents, and compliance documentation (where terms and definitions are often mandated from regulatory guidelines and/or standards). These terms often have more than one meaning or there may be multiple terms with the same or similar meanings to support multiple perspectives (from stakeholders). These additional meanings are often accompanied by specific context associated with a given perspective, so this context needs to be captured using a disciplined process alongside the semantics to support the business perspectives (known today and needed for tomorrow).

As the business vocabulary emerges, there is a need to create a higher-level depiction to represent the business using subject areas, extracted from the above-mentioned artifacts. The subject areas are typically arranged in an information map, which is used as a foundation to guide the creation / refinement of the information architecture. This reference model (if you will) needs to align to the capabilities, the business vocabulary (should be able to find a home for all the business terms and semantics), and processes (where all the resources / roles can be associated with one or more subject areas). The relationships between these subject areas depict the flow and interdependencies at an organization level.

## **Consider Points of View on Information**

It is tempting to conclude that the term “Customer” has a simple, common definition. The following shows that this is not the case:

- A customer is someone who may purchase my product (purchase agreement management point of view).
- A customer is someone who may fail to pay bills on time (treasurer/collections point of view).
- A customer is someone who likes my product (LoB executive point of view).
- A customer is someone who receives a product (fulfillment point of view).

These different points of view typically correlate with specific capabilities of the business. Another example is where functional areas within an organization refer to Product with different perspectives:

- External Sales may reference product as finished goods with an SKU

- Marketing may reference product as one or more finished goods (a bundle perhaps)
- R&D may refer to product as a concept under development (which itself may have a Product ID)
- Supply Chain Operations may refer to partially finished goods as both product and material (and it's possible to have both an item number and a product ID)

One more example, which illustrates some the complexity with vocabulary and semantics, is when different terms are used to refer to the “same thing”. Take for example the terms Customer and Account. For Sales Operations, they are interchangeable, but for Supply Chain Operations and Customer Support, they may be very different. The challenge (and opportunity) with addressing these perspectives is to accommodate them (supporting the existing culture, operational processes, and systems) in place while enabling a consistent view of Customer or Product (just 2 examples) across the organization.

These perspectives are important to capture and carry forward as we map information concepts (objects, terms, and relationships) to capabilities (initially), processes, systems, and knowledge (eventually). It is useful to have a common definition of terms (like Customer or Product) with the ability to support these different points of view as it reinforces the notion that the terms in the Business Glossary have relevancy to most (if not all) stakeholders and have varying levels of importance at different points in time (this is especially true where terms are associated with lifecycles).

The common approach to addressing these perspectives is to use a group-based approach to agree upon a homogenized definition of a given term. This is problematic since the contextual relevance for a given perspective is lost in translation. A better approach is to preserve this contextual relevance within the business glossary where it can be captured (over time) as these perspectives will grow and mature as the understanding of stakeholders' needs are uncovered during projects (organic approach).

This is where a business glossary of terms can play a significant role. The glossary is where these core terms (object classes and attributes for Product and Customer) are defined and it is also where the contextual terms (additional object classes and attributes) are defined, with rule sets that capture the relationships between these terms and attributes to support the semantic relationships inherent in these perspectives.

Many organizations do not use the term customer or use it in a very different way than what is defined above. Alternatives commonly sought include Member, Owner, Holder, Patient, Licensee, or Constituent. For this reason, we use labels on concepts, allowing the concept to be identified by several terms or phrases.

The capture, warehousing, and extraction (usage) of the vocabulary needs to be managed with a set of disciplined processes (via Governance). Information and understanding starts with a solid BA MM and is extended with a well-managed Business Glossary that supports the Meta Data for organization.

## **Identifying Business Objects & Building a Conceptual View**

Business objects represent higher-level meta-types of terms and information objects that are typically derived from the BA MM using a top-down approach. These concepts are useful for developing a reference model for the business to align portfolios of assets (includes resources) across the organization with the core maps for guiding value creation (value streams, capabilities, etc.) to optimize value creation via the models used to drive execution and operationalize value delivery (projects, processes, operations, supply chain, GRC, product, service, information, integration, etc.).

Examples of business objects include:

- Customer
  - Derived from Stakeholder via the BA MM
  - Covers direct customers (retailers, 3<sup>rd</sup> party value added resellers / mfgs), indirect customers (warehousing distributors, wholesalers), and consumers who buy and consumer products and services.
  - Includes but is not limited to information objects (and their attributes) for Stakeholder, Customer, Party, Distributor, Consumer, Persona, Location, Profile, Product, Order, ...
- Service
  - Derived from Product via the BA MM
  - Covers external services to 3<sup>rd</sup> parties (typically customers), where value is delivered via operational processes.
  - Includes but is not limited to information objects (and their attributes) for Stakeholder, Customer, Party, Product, Process, Compliance, Order...

## **Aligning Business Objects with Information Objects**

This is where we identify the alignment (first) and clarify the nature of the relationships (second) between business and information objects. This is a critical step to validate the completeness of the inventory of business and information objects. This “matrix” also helps identify incomplete lineage between business objects and information objects (e.g. there should not be orphan information objects). The matrix also helps identify dependencies

between information objects (required to support business objects). This serves as a quick reference card for the business to guide information management and data governance efforts.

### **Drafting the Information Map (Conceptual View of the Business)**

The Information Map is a conceptual view of the business and is comprised of business and information objects and their relationships (high level only) to each other. This conceptual view provides a business friendly (non-technical) and consistent understanding of the organization and how it is put together. This conceptual model is fairly static and provides the baseline from which all other maps (value stream, capability, process, and knowledge) are put together. The conceptual view also grounds and bounds all of the models (objects, process, rules, information, operational, technical, etc.) used to guide the design and implementation for the information architecture (including integration and interoperability), business operations, application architecture, and the business architecture.

### **Drafting the Information Metadata (Object View of the Business)**

The terms and relationships that stem from the conceptual view are used to form the information objects that describe the WHAT, WHY, WHERE, HOW, & WHEN across the business and across the (one or more) value chain(s) where the business participates. These information objects are captured in a formalized set of taxonomies within a technical structure called Metadata. This information metadata is used to support a common understanding of how things are related across every area of the business regardless of time (as things change and may look and work differently in the past, present, and future states). This common understanding is captured using the core information objects that form the essence of the business (Customer, Employee, Vendor, Product, Location, Process, Service, Capability, Organization). These information objects are used across the organization and the value chain(s) and are complimented with contextual information objects (Preference, Profile, Persona, Lifecycle, Event, Value, Agreement, Concept...) used to support the many perspectives that are known today and that will be discovered in the future.

The core information objects are fairly stable taxonomies once they are defined. However, the contextual information objects tend to be grow quite rapidly as more external information is brought into the organization for consumption (think Big Data). These contextual taxonomies provide a flexible and adaptable capability for extending understanding for Big Data and for “Deep” Analytics.

The role of the business architect in helping for form this Metadata is critical since these information objects need to support the business needs and must support and enable the maps and models required to plan, create, and deliver value for stakeholders (today and tomorrow).



## Socializing & Validating the Information Maps

Throughout the development of the business vocabulary, semantics, and the information maps (conceptual, logical, and metadata), there is an ongoing dialogue with the subject matter experts (SMEs) in the business, the business architect, and the information architect / modeler. The validation of these business / information artifacts is an ongoing and iterative process. For example: As Capability Maps get defined and refined, updates are made to the vocabulary, the semantics, and the models. Similar change cycles follow during project work as new scenarios are uncovered, processes are defined and/or improved, and value stream gaps are addressed.

The formation of the teams used to socialize the Information maps and related artifacts as well as the governance processes will be discussed further in the next set of sub-sections. A disciplined process is typically put in place to track the lineage (traceability) between the information and artifacts provided by the business SMEs and the information assets associated with the Information Maps. The process assures the dialogues are properly documented and all changes to the information artifacts involved in the creation of Information Maps or are derived from the Information Maps. This level of traceability typically has roots in quality assurance with crossover into configuration management, which provides end-2-end auditability throughout the information supply chain and related lifecycles for these artifacts.

The validation of the Information Map and the related artifacts (upstream and downstream) is performed during the iterative reviews after one or more “sets” of changes are made to the documents used to capture the traceability.

## Validation Guidelines for the Information Map

The following validation guidelines apply to working with information maps.

### ***1. Determine that an information concept is a business information concept.***

All information concepts included in the mapping should be business concepts and not concepts that have been artificially created to accommodate a particular technical or data modeling implementation.

### ***2. Ensure that an information asset has a corresponding capability.***

If an information concept is used in the business, it will likely have a close correspondence to a business capability. We demonstrated this notion in figure 2.5.7.

### ***3. Validate that a concept is generalized across the business where appropriate.***

An insurance group and a financial management group, for example, would need to concur on the concept of Account versus Policy. One organization used the term

Solution as a way to compromise across divisions. The definition incorporated the definitions of Policy and Account and allowed the use of these terms as a more specific term than Solution.

***4. Inspect initiating stakeholders for each value stream to ensure that all major stakeholder categories are included.***

The guideline here is to roll stakeholder names up to reflect common stakeholder categories such as Customer, Supplier, Employee, Agent, and similar concepts. This rollup is noted in the information map as an "is a kind of" linkage.

***5. Ensure that each business information asset is socialized and validated with a representative cross-section of the business.***

This concept, common to all business architecture mapping exercises, ensures that the concepts, terms, and definitions are indeed business terminology that is generally agreed upon within the business.

## **Guiding the Design for Logical Model (and drafting of Business Rules)**

As stated earlier, the Logical Model for the information architecture must be aligned with the Conceptual Model, where levels of detail are created using Information Objects. The Logical Model provides an object-based view for all domains and should address any / all scenarios that the organization needs to address. These scenarios cover “use cases” that may impact the processes, systems, and underlying information at the transaction level, operational level, and at the decision-making level (via reports, KPIs, analytics, or big data).

## **Aligning Information Maps with Other Architecture Domains**

The Information Maps (and related information assets discussed above) are used to guide integration and support interoperability of systems, processes, business units, and organizations. The data objects, taxonomies, and ontologies can be directly applied to canonical mappings between providers and consumers of data, information, and knowledge (includes content and patterns built atop unstructured data & information). Increasingly, organizations are leveraging automated processes (services) to handle transactions and information exchange between their own systems and with 3<sup>rd</sup> party owned systems. Many of these systems include those owned by government agencies (think compliance and regulatory oversight) as well as value-added partners in the product supply chain.

The systems, automated processes, human interfaces (websites, mobile channels), and workflow/event management capabilities are all reliant on consistent views of data, information and the explicit relationships between data and information objects. The flexibility of the application architecture is often bound by the vendors and 3<sup>rd</sup> party systems used to

support the enterprise applications (on premise and off premise). Aligning the data, information, and knowledge across these systems requires a flexible and agile design for the integration layer, information / analytic layer, and metadata management. Achieving this level of maturity requires collaboration across the architecture disciplines and needs guidance from the business architect at each step along the maturity curve. Keeping the business needs in focus while balancing the short –term (project needs) against the long-term (architecture maturity needs) requires patience, ongoing communications, and building trust amongst the project / portfolio teams and the technology teams.

## **Information Governance**

There are 2 primary goals for governing the information supply chain: The first goal is to ensure that the data, information, and knowledge that “flows” through the supply chain (river and all of its tributaries) is high quality (to be defined). The second goal is to ensure that that data, information, and knowledge has not been compromised or accessed inappropriately. These goals are typically referred to as Data Quality (DQ) and Data Security (DS), but are not constrained to the “data” since data is transformed into information and knowledge as it flows downstream. Governance is more challenging with respect to the information supply chain since it is difficult to determine statefulness (condition of the data at any one point in time) as is constantly flowing. This sub-section will discuss several governance models that may be useful to consider and the role that business architecture can play in guiding the business and technology (including the other architecture disciplines) through a more effective approach for establishing and managing DQ & DS for the information supply chain.

### **Data Quality (integrity)**

At the end of the day, information is used to make decisions and knowledge is used to inform the decisions that are made. The information supply chain is built on trust and that trust needs to extend to every part of the supply chain (from where data is sourced, how it is profiled and cleansed, where it is warehoused, how it is transformed, to when and how it is extracted and reused, and even how and when it is archived and destroyed). Data Quality issues can impact every part of an organization and can result in misstatements on financial documents, regulatory violations or non-compliance, over or under funding reserves, additional paperwork, fines, and other challenges. Identifying the source of a data quality problem is a non-trivial exercise. It requires skilled resources from many parts of an organization to determine where in the information supply chain the issue occurred and why it occurred. Many organizations who have robust programs in Lean or Six Sigma have extended these processes into their data governance programs. Data Governance provides a more proactive approach towards identifying and preventing data issues across the information supply chain.

### **Data Security (Confidentiality, Integrity, Availability)**

Due to the ubiquitous nature of information and integration of processes and systems within organizations, between organizations, and throughout value chains, the challenge of securing data throughout the information supply chain continues to grow more complex. The use of firewalls and access control lists (ACLs) worked fine when security focused on the “four walls” of an organization where little if any information was passed between 3<sup>rd</sup> parties. The proliferation of 3<sup>rd</sup> parties who provide / consume data, information, and knowledge with the organization’s information supply chain suggests the need for more adaptable and mobile

security capabilities to keep pace with the increasing exposure. Think about securing the water supply as it flows through all the pipes, gets stored in reservoirs, and is combined with other materials, which are eventually consumed in the food supply (and drug supplies). Each point along the supply chain needs to be measured and controlled as well the processes and systems that interact with the data, information, and knowledge.

The unique challenge with information is that it is a reusable commodity / asset, so ensuring its security requires knowledge about the other parties that directly and indirectly need to gain access for creating, reading, updating, deleting, and searching (CRUDS). In addition, the stewardship of data throughout its lifecycle requires collaboration amongst all of the parties (individuals, departments, functional areas, 3<sup>rd</sup> parties) who participate in the supply chain. The following sub-sections will discuss dimensions of governance within an organization, but the concepts can be extended to multiple parties beyond the organization.

## **Principles of Information Governance**

Before exploring governance models, it's helpful to review some of the principles that are commonly promoted by leading organizations. The principles that follow are pulled from the Data Governance Guiding Principles as published from the Data Governance Institute. These principles provide a solid starting point for establishing a successful Data Governance and Stewardship program, defining and implementing operational processes for data management, and guiding projects to improve data quality and data security. They are the principles that help stakeholders come together to resolve the types of data-related conflicts that are inherent in every organization.

### **Integrity**

Data Governance participants will practice integrity with their dealings with each other; they will be truthful and forthcoming when discussing drivers, constraints, options, and impacts for data-related decisions.

### **Transparency**

Data Governance and Stewardship processes will exhibit transparency; it should be clear to all participants and auditors how and when data-related decisions and controls were introduced into the processes.

### **Auditability**

Data-related decisions, processes, and controls subject to Data Governance will be auditable; they will be accompanied by documentation to support compliance-based and operational auditing requirements.

### **Accountability**

Data Governance will define accountabilities for cross-functional data-related decisions, processes, and controls.

### **Stewardship**

Data Governance will define accountabilities for stewardship activities that are the responsibilities of individual contributors, as well as accountabilities for groups of Data Stewards.

### **Checks-and-Balances**

Data Governance will define accountabilities in a manner that introduces checks-and-balances between business and technology teams as well as between those who create/collect information, those who manage it, those who use it, and those who introduce standards and compliance requirements.

### **Standardization**

Data Governance will introduce and support standardization of enterprise data.

### **Change Management**

Data Governance will support proactive and reactive Change Management activities for reference data values and the structure/use of master data and metadata.

## **Exploring Governance Models**

Establishing governance around data, information, and knowledge can take anywhere from a few months to over a year depending on the culture and readiness of the organization. First and foremost, the business needs to recognize the value of information as an asset. Then, there needs to be sufficient momentum for investing the time and resources to establish a governance framework (complete with processes, controls, measurement, and continuous improvement). That momentum can be triggered by a change in business direction (potential value to an existing or new line of business that relies on information), recognition of exposure (and impact) from a data issue (quality and/or security).

Governance typically starts with the formation of an executive committee to identify ownership around DQ & DS for one or more core subject areas (Customer, Product, Vendor, Employee, ...). The committee may assume ownership (initially) to implement controls and stabilize one or more sets of issues. Often times, the committee will identify one or more senior leaders in functional areas to be accountable (think RACI) for one or more of the core subject areas.

Governance can also be established as an add-on to existing governance models covering core value streams for the organization, operations, processes, systems / technology platforms, core

information sources / warehouses, and even functions across the organization. Each organization needs to determine what works best for their capabilities, resources, and culture.

Ultimately, governance is a business owned and driven initiative and needs to align with existing governance models (Technology, Compliance, Product Portfolio, Project Portfolio, Process, Risk, Security, etc.). The Business Architect can play a key role in helping to align these governance models and integrate them to reduce the overhead on leadership while streamlining the deployment of resources to manage the operations associated with these governance processes, address issues, and drive continuous improvement.

## **Executive Oversight**

Typically, information governance is setup using a 2 or 3-tier model where the top tier is the decision-making body that defines and enforces policies that guide activities around the information supply chain (including all the lifecycles and portfolios of assets). The executive oversight is where the buck stops for DQ & DS. If there is a problem (or opportunity) with information that needs to be addressed, the action plans are brought before this executive team/committee for review / approval and prioritization. The executive oversight also needs to draft and enforce policies to ensure DQ & DS across the information supply chain. These policies can take time to develop and mature, so this requires a patient and iterative process for all who are involved. The key for guiding effective policy development and enforcement is measurement. The executive team cannot be expected to make informed decisions without a well thought out approach for measuring DQ & DS. The business architect needs to socialize / promote the need and discipline for measurement amongst the executive team so that the technology resources can be allocated to build and support these measures.

Note: The reason for the integrated governance (mentioned above) is that these decisions can have significant impacts throughout the organization so they cannot be made in a vacuum nor is the prioritization around any approved actions (initiatives) a simple matter.

## **Operational Management**

The executive team relies upon an operational team of resources across the organization (and value chain in more mature settings) to monitor DQ & DS, manage the data-to-day issues, and provide business level expertise (including impact analysis) in projects, and offer guidance to the executive team (and functional areas across the organization). The operational team can be setup using a variety of approaches:

**A matrix approach** (with data subject owners who manage the end-2-end lifecycle)

Pros: clear ownership of data at any point in its lifecycle

Cons: requires re-allocation of resources (some of whom are highly skilled)

**Virtual teams** that come together for projects and to address issues

Pros: ad-hoc, agile approach, which is an extension of existing activities with light process and lower overhead

Cons: more informal handling of issues at times, can lead to inconsistencies, and gaps in continuity for initiatives and/or issue management

**Functional area (system-based) teams** that have primary ownership over one or more subject areas where most of the data flows through the systems and/or processes for which they are responsible

Pros: tight control over all processes, systems, and information assets within their ownership

Cons: may lead to gaps in control as data flows beyond the control of their functional processes and/or systems

Every organization approaches this with different levels of understanding and commitment. This is not an optional business investment. This needs to be carefully planned, implemented, and controlled. The business architect needs to coach and mentor the business SMEs as to the importance of building a sustainable capability. This business will need a steady, continuous reinforcement, and a solid communication program to ensure the success of this capability. This is less about technology and more about change management and designing for sustainability (it's a marathon not a sprint).

## Organizing the Information Mapping Team

The information mapping team is largely made up of the same roles as the capability mapping team described in the *BIZBOK® Guide* section 2.2. The only variation in regards to the information mapping team is that it would involve two additional individuals to work with the team as a sounding board.

- Information Architects specialize in the knowledge, information and wisdom needs of the organization. They are trained in techniques for modeling business information and linking it to the other viewpoints of the business architecture.
- Data Architects are experts on data modeling and organizing enterprise data for a variety of scenarios. They do not have the business knowledge or access to business professionals required to create the information map from a business perspective.



The data architect serves as an advisor/mentor because this individual understands many of the issues encountered in dealing with information in the enterprise today, including knowledge of data held by the organization, how that data is collected, and its governance and security policies.

There are individuals in some organizations who can further augment the information mapping exercise. These individuals are often called data stewards and data champions. Their roles are defined as follows.

- Data Stewards live within a subject area typically defined by a common capability. For example, a claims team may have a resident or rotating steward for common claims processes and claims related information. These individuals may already have completed some of the groundwork required to align terms and definitions across a given area, which is typically defined by a level one capability. Stewards are responsible for the quality of the data they represent.
- Data Champions act as conduits between business and IT and typically are experienced in the subject area as well as having knowledge of who would be engaged to ensure that the all parties have bought into various terms and definitions.

Steward and champion roles are not essential, but if they exist within an organization already then the team should leverage their knowledge accordingly.

### Aligning Priorities for Information Governance

To be drafted...

### Leveraging BA Heat Maps to Guide Information Governance

The information map can be heat mapped to indicate information that is unavailable or of poor quality. However, an examination of the associated heat-mapped capabilities will be more valuable as it indicates the degree to which the functioning of the capability is inhibited by an information deficit. One should be careful to consider if the deficit is caused by lack of information, or by the inability of the employees to make efficient use of the available information, as this will require a different kind of business transformation to remedy the deficiency. The latter is a deficiency of the capability rather than a deficiency of the information.

It will often be the case that information desired by the business, its partners, and its customers will not be available because it is not collected or because it is not known how to obtain the information. These cases are particularly challenging for the business architect. Information,

known to be useful to a business may not be available because collecting it is too expensive or is infeasible because the information is tacit or cultural. In such cases, the business architect, working with the stakeholders of the business architecture, may be able to design an information proxy that can be collected and which is thought to exhibit the same behavior as the ideal information.

## **Measurement and Continuous Improvement**

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